

4a	<u>work done per unit positive charge</u> in moving a <u>point charge</u> from <u>infinity to that point</u>	B1
bi	p.d. = 15 V	A1
bii	change in potential $\Delta V = 25 - 5 = 20 \text{ V}$ work done = $q\Delta V = 5 \times 10^{-6} \times 20$ $= 1.0 \times 10^{-4} \text{ J}$	C1 A1
biii	The <u>potential increase</u> positively as the charge move <u>from B to E.</u> Since it is a <u>positive charge</u> , <u>work has to be done by external agent</u> to move it to a more positive potential line.	M1 A1
ci	electric field strength at a point is equal to the <u>negative potential gradient</u> at that point	B1
cii	<u>towards the +20 V</u> equipotential line and approximately <u>perpendicular</u> at the point.	B1
ciii	move away to the right / towards lower potential / away from higher potential velocity increases at increasing rate / acceleration <i>Learning point:</i> <i>initial acceleration is small, acceleration is (much) larger beyond the +20 V line</i>	B1 B1
	Total:	10